

# LEGAL EASE

AI-Powered Legal Document Generator

## Project Development Document

**Project Overview:** LegalEase is an AI-powered application designed to help users generate structured legal-document drafts from simple user inputs. The system collects the required information, processes it using AI, and produces a formatted document that can be reviewed and edited by the user. The project is intended as an academic prototype and does not replace professional legal advice.

### Project Phases

| No. | Phase                          | Main Activities                                                                                                                           |
|-----|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | Brainstorming & Ideation Phase | Identify the problem, discuss possible solutions, define the LegalEase concept, and identify target users.                                |
| 2   | Requirement Analysis Phase     | Collect functional and non-functional requirements, identify user inputs, define AI requirements, and determine the legal document types. |
| 3   | Project Design Phase           | Design the system architecture, user interface, input forms, AI processing flow, document-generation logic, and database structure.       |
| 4   | Project Planning Phase         | Prepare the development schedule, divide tasks into modules, select technologies, estimate resources, and set milestones.                 |
| 5   | Project Development Phase      | Develop the interface, backend, AI prompt/workflow, document generator, validation features, and integrate with a legal database.         |
| 6   | Project Testing Phase          | Test inputs, document generation, formatting, error handling, usability, and system performance; fix bugs and optimize.                   |
| 7   | Project Documentation Phase    | Prepare project report, system description, screenshots, module explanations, testing results, limitations, and future work.              |
| 8   | Project Demonstration Phase    | Present the problem, explain the solution, demonstrate the application workflow, show generated documents, and answer questions.          |

## 1. Brainstorming & Ideation Phase

- Problem identified: creating legal-document drafts can be difficult for users who are unfamiliar with legal formats.
- Proposed solution: an AI-assisted document generator that converts user-provided details into a structured draft.
- Possible document categories can include agreements, notices, applications, declarations, and other predefined templates.
- Define the target users, project scope, expected benefits, and limitations.

## 2. Requirement Analysis Phase

- Functional requirements: user input collection, document-type selection, AI-assisted content generation, preview, editing, and export.
- Non-functional requirements: simple interface, reasonable response time, input validation, reliability, and basic privacy considerations.
- The system should clearly distinguish generated drafts from professional legal advice.
- Required inputs and expected outputs are documented before development.

## 3. Project Design Phase

- Design the overall workflow: User Input → Validation → AI Processing → Draft Generation → Preview/Edit → Export.
- Create UI screens such as Home, Document Selection, Input Form, Generated Draft, and Download/Export.
- Design the backend modules and the data flow between the interface, AI service, and document generator.
- Prepare the database structure if user/project data needs to be stored.

## 4. Project Planning Phase

- Break the project into manageable modules and assign responsibilities.
- Prepare a development timeline covering design, coding, integration, testing, documentation, and demonstration.
- Select suitable technologies based on project requirements and available resources.
- Define milestones and expected deliverables for each stage.

## 5. Project Development Phase

- Build the frontend and input forms.
- Implement backend logic and connect the required AI service or AI workflow.
- Create prompts/templates for generating structured legal-document drafts.
- Implement document formatting and export functionality.

- Integrate all modules and verify that the complete workflow operates correctly.

## 6. Project Testing Phase

- Perform functional testing using different valid and invalid inputs.
- Check whether the generated document contains the requested information and follows the selected structure.
- Test validation, empty fields, unexpected input, formatting, and export functions.
- Record defects, correct them, and repeat testing after fixes.

## 7. Project Documentation Phase

- Prepare the project abstract, objectives, problem statement, methodology, modules, system requirements, and workflow.
- Include screenshots of important screens and sample generated output.
- Document test cases, results, limitations, and future enhancements.
- Add user instructions explaining how to enter information and generate a document.

## 8. Project Demonstration Phase

- Introduce the problem and explain the purpose of LegalEase.
- Show the application from document selection to final generated draft.
- Explain the major modules and technologies used.
- Demonstrate validation, preview/editing, and export features.
- Conclude with project benefits, limitations, and possible future improvements.

## Expected Outcome

The completed LegalEase prototype should provide a simple workflow for generating well-structured legal-document drafts from user inputs. The application should reduce manual formatting effort while keeping the user in control of reviewing and editing the generated content before use.

## Future Enhancements

- Support for more document categories and reusable templates.
- Multiple language support, including regional languages.
- Improved document formatting and downloadable file formats.
- User accounts and saved document history, subject to appropriate privacy safeguards.
- Human/legal-professional review workflow for higher-stakes use cases.

**Note:** LegalEase is an academic project concept. AI-generated content should be reviewed carefully and should not be treated as a substitute for advice from a qualified legal professional.